Supplementary Information

Four-Corner Spectral Tuning for Quadratic ADMM with Application to Diffeomorphic Image Registration

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1 Purpose and notation

This supplement develops the optional noncommuting rate certificates, their proofs, scaling diagnostics, and implementation details. These constructions are not used to choose the deployed four-corner parameters. We abbreviate the alternating direction method of multipliers as ADMM and its over-relaxed form as oADMM. For symmetric positive semidefinite $H, G \in \mathbb{R}^{dn\times dn}$ and $\rho > 0$, define

$$C_A(\rho) = (A - \rho \mathrm{I})(A + \rho \mathrm{I})^{-1},$$

for any symmetric positive semidefinite A, and abbreviate

$$C_H = (H - \rho \mathrm{I})(H + \rho \mathrm{I})^{-1}, \qquad C_G = (G - \rho \mathrm{I})(G + \rho \mathrm{I})^{-1}.$$

The reduced oADMM error operator is

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right)\mathrm{I} + \frac{\alpha}{2} C_H C_G, \qquad 0 < \alpha \leq 2. \tag{1}$$

Let nonoverlapping patches $\{\Omega_r\}$ partition the full grid, and define the full-size direct-sum surrogate

$$\widetilde{H} = \bigoplus_r (I_{|\Omega_r|} \otimes H_r), \qquad \widetilde{G} = \bigoplus_r (G_r \otimes I_d),$$

where each H_r and G_r is symmetric positive semidefinite. Let

$$\delta_H = \left\| H - \widetilde{H} \right\|_2, \qquad \delta_G = \left\| G - \widetilde{G} \right\|_2, \qquad \delta = \delta_H + \delta_G.$$

The two surrogate operators commute blockwise. If the symbol endpoints used in the modal maximization are attained, its exact frozen-patch envelope satisfies $U_{\text{patch}} = \left\| \widetilde{E}_{\rho,\alpha} \right\|_2$. The notation "exact" always refers to this stated frozen separable model; the full variable-coefficient registration operator is generally noncommuting.

2 Additive perturbation certificate

2.1 Cayley Lipschitz estimate

Lemma 2.1 (Cayley Lipschitz bound)**.** *For symmetric positive semidefinite A, B and $\rho > 0$,*

$$\left\| C_A(\rho) - C_B(\rho) \right\|_2 \leq \frac{2}{\rho} \left\| A - B \right\|_2 . \tag{2}$$

Proof. Writing $C_A = \mathrm{I} - 2\rho(A+\rho\mathrm{I})^{-1}$ and using the resolvent identity gives

$$\begin{aligned} C_A - C_B &= 2\rho\left[(B+\rho\mathrm{I})^{-1} - (A+\rho\mathrm{I})^{-1}\right] \\ &= 2\rho(A+\rho\mathrm{I})^{-1}(A-B)(B+\rho\mathrm{I})^{-1}. \end{aligned}$$

Both resolvent norms are at most ρ^{-1}, proving (2). $\square$

Theorem 2.2 (Additive frozen-patch certificate)**.** *For the direct-sum surrogate defined above and the full variable-coefficient iteration of the same dimension,*

$$\varrho(E_{\rho,\alpha}) \le U_{\text{patch}}(\rho,\alpha) + \frac{\alpha\delta}{\rho}. \tag{3}$$

No commutativity assumption on H and G is required.

Proof. Since every Cayley transform of a positive semidefinite matrix has norm at most one,

$$\begin{aligned} \left\|C_H C_G - C_{\widetilde{H}} C_{\widetilde{G}}\right\|_2 &\le \left\|(C_H - C_{\widetilde{H}})C_G\right\|_2 \\ &\quad + \left\|C_{\widetilde{H}}(C_G - C_{\widetilde{G}})\right\|_2 \\ &\le \frac{2}{\rho}(\delta_H + \delta_G). \end{aligned}$$

Apply (1), the triangle inequality, and normality of the separable direct-sum operator, whose operator norm equals U_{patch}. Any cross-patch couplings removed from the regularizer are included in δ_G. $\square$

Corollary 2.3 (Certified linear rate)**.** *If the right-hand side of (3) is below one, then the full linearized oADMM iteration converges linearly with asymptotic factor no larger than that value.*

2.2 Certificate-aware legacy objective

For standard ADMM, all scalar frozen-modal factors are nonnegative. Let $\mathcal{C}$ denote the finite set of attained frozen-model curvature–symbol corners. A certificate-aware diagnostic objective is therefore

$$\mathcal{U}_1(\rho) = \max_{c\in\mathcal{C}} \frac{\rho^2 + h_c g_c}{(\rho+h_c)(\rho+g_c)} + \frac{\delta}{\rho}. \tag{4}$$

For two corner branches $c_1 = (h_1, g_1)$ and $c_2 = (h_2, g_2)$, equality after denominator clearing is

$$\rho(A_{12}\rho^2 + B_{12}) = 0,$$

where

$$A_{12} = (h_2 + g_2) - (h_1 + g_1), \qquad B_{12} = g_1 g_2 (h_1 - h_2) + h_1 h_2 (g_1 - g_2).$$

On an interval where $c = (h, g)$ is active, stationarity of (4) gives

$$(h+g)\rho^2(\rho^2 - hg) - \delta(\rho+h)^2(\rho+g)^2 = 0. \tag{5}$$

Thus the legacy standard-ADMM bound can also be minimized over a finite set of endpoints, branch intersections, and active quartic roots. This diagnostic is not the deployed four-corner selector: the additive term often moves its minimizer toward an unhelpful boundary.

For an admissible interval $[\alpha_-, \alpha_+] \subset (0, 2]$, let a and b be the active minimum and maximum frozen-modal θ values at the selected penalty. The certificate-aware relaxation objective is

$$F(\alpha) = \max\{|1 - \alpha + \alpha a|,\ |1 - \alpha + \alpha b|\} + \frac{\alpha\delta}{\rho}.$$

It is convex and piecewise linear. A minimizer belongs to

$$\left\{\alpha_-, \alpha_+, \frac{1}{1-a}, \frac{1}{1-b}, \frac{2}{2-a-b}\right\} \cap [\alpha_-, \alpha_+].$$

Only finite, defined breakpoints are retained. If $b = 1$, the modal maximum is identically one on $(0, 2]$, so $F(\alpha) = 1 + \alpha\delta/\rho$. Its minimizer is α_- when $\delta > 0$, and every admissible α is optimal when $\delta = 0$.

3 Exact-regularizer block structure

The additive certificate replaces both operators. Sharper bounds retain the regularizer exactly. Let

$$G = G_0 \otimes I_d, \qquad K_\rho = (G_0 - \rho I_n)(G_0 + \rho I_n)^{-1},$$

so $C_G = K_\rho \otimes I_d$. Write

$$C_H = \operatorname{diag}(C_1, \ldots, C_n), \qquad C_i = (H_i - \rho I_d)(H_i + \rho I_d)^{-1}.$$

Then the (i, j) block of $M := C_H C_G$ is

$$M_{ij} = (K_\rho)_{ij} C_i. \tag{6}$$

Although K_ρ and every C_i are symmetric, M is generally nonnormal because the C_i vary.

4 Block-Gershgorin disks

Theorem 4.1 (Block-Gershgorin enclosure)**.** *Every eigenvalue of $M = C_H C_G$ lies in a disk*

$$|z - (K_\rho)_{ii} c_{i,r}| \le \|C_i\|_2 \sum_{j \ne i} |(K_\rho)_{ij}| \tag{7}$$

for some pixel i and eigenvalue $c_{i,r}$ of C_i. Consequently

$$\begin{aligned} U_{\mathrm{BG}}(\rho, \alpha) = \max_{i,r} \Big[& \left|1 - \frac{\alpha}{2} + \frac{\alpha}{2}(K_\rho)_{ii} c_{i,r}\right| \\ & + \frac{\alpha}{2} \|C_i\|_2 \sum_{j \ne i} |(K_\rho)_{ij}| \Big] \end{aligned} \tag{8}$$

is a rigorous upper bound on $\varrho(E_{\rho,\alpha})$.

Proof. Let $Mx = zx$ and choose a block x_i of maximum Euclidean norm. The ith block equation is

$$(zI_d - (K_\rho)_{ii} C_i)\, x_i = \sum_{j \ne i} (K_\rho)_{ij} C_i x_j.$$

The diagonal block is normal, so its smallest singular value is the distance from z to its spectrum. Taking norms, using $\|x_j\| \le \|x_i\|$, and cancelling $\|x_i\|$ gives (7). The affine map from M to $E_{\rho,\alpha}$ maps each disk to the bound (8). □

5 Signed rectangle and commutator information

Decompose

$$S = \frac{M + M^\top}{2}, \qquad Q = \frac{M - M^\top}{2}.$$

For $i \neq j$,

$$S_{ij} = (K_\rho)_{ij} \frac{C_i + C_j}{2}, \qquad Q_{ij} = (K_\rho)_{ij} \frac{C_i - C_j}{2}.$$

Define zero-diagonal symmetric comparison matrices by

$$B^S_{ij} = |(K_\rho)_{ij}| \, \|(C_i + C_j)/2\|_2 \,, \qquad B^Q_{ij} = |(K_\rho)_{ij}| \, \|(C_i - C_j)/2\|_2 \,, \tag{9}$$

and let d_i^- and d_i^+ be the extreme eigenvalues of the diagonal block $(K_\rho)_{ii} C_i$. The block-row quantities are

$$m_\rho^{\mathrm{row}} = \min_i \left[d_i^- - \sum_{j \neq i} B^S_{ij} \right], \tag{10}$$

$$M_\rho^{\mathrm{row}} = \max_i \left[d_i^+ + \sum_{j \neq i} B^S_{ij} \right], \tag{11}$$

$$\eta_\rho^{\mathrm{row}} = \max_i \sum_{j \neq i} B^Q_{ij}. \tag{12}$$

Theorem 5.1 (Commutator-aware signed rectangle)**.** *With $W(M) = \{x^* M x : \|x\|_2 = 1\}$ denoting the numerical range,*

$$W(M) \subset [m_\rho^{\mathrm{row}}, M_\rho^{\mathrm{row}}] + \mathrm{i}[-\eta_\rho^{\mathrm{row}}, \eta_\rho^{\mathrm{row}}].$$

Therefore

$$\varrho(E_{\rho,\alpha}) \leq U_{\mathrm{rect}} := \max_{x \in \{m_\rho^{\mathrm{row}}, M_\rho^{\mathrm{row}}\}} \sqrt{\left(1 - \frac{\alpha}{2} + \frac{\alpha x}{2}\right)^2 + \left(\frac{\alpha \eta_\rho^{\mathrm{row}}}{2}\right)^2}. \tag{13}$$

Proof. For any normalized complex vector x, set $z = x^* M x$. Then

$$\operatorname{Re} z = x^* S x, \qquad \mathrm{i} \operatorname{Im} z = x^* Q x.$$

Hermitian block-Gershgorin bounds applied to S put the first quantity in the stated real interval. The symmetric matrix of block norms bounds $\|Q\|_2$ by its maximum row sum, giving $|\operatorname{Im} z| \leq \eta_\rho^{\mathrm{row}}$. Applying the affine map in (1) and maximizing the Euclidean modulus over a rectangle gives (13); convexity puts the real-coordinate maximum at an endpoint. □

For fixed ρ, the squared expressions under the maximum in (13) are quadratics in α. Their minimum is attained at an admissible interval endpoint, a stationary point of either quadratic, or their intersection

$$\alpha = \frac{4}{2 - m_\rho^{\mathrm{row}} - M_\rho^{\mathrm{row}}}$$

when admissible. This finite conditional selection is useful for analyzing a certificate but is not used in one-shot registration tuning.

6 Perron comparison rectangle

The row maxima can be replaced by global scalar eigenproblems.

Theorem 6.1 (Perron comparison)**.** *The signed rectangle theorem remains valid with*

$$\begin{aligned} m_\rho &= \lambda_{\min}\left(\operatorname{diag}(d^-) - B^S\right), \\ M_\rho &= \lambda_{\max}\left(\operatorname{diag}(d^+) + B^S\right), \\ \eta_\rho &= \varrho(B^Q). \end{aligned} \tag{14}$$

These bounds are no weaker than their block-row counterparts. Denote the resulting value of (13) *by* U_{cmp}*.*

Proof. Partition a complex block vector x and set $y_i = \|x_i\|_2$. For the symmetric part,

$$x^* S x \le \sum_i d_i^+ y_i^2 + 2\sum_{i<j} B_{ij}^S y_i y_j = y^\top (\operatorname{diag}(d^+) + B^S) y.$$

The analogous lower estimate uses $\operatorname{diag}(d^-) - B^S$. Rayleigh–Ritz yields the first two bounds. Blockwise comparison for the skew part gives

$$\|Qx\|_2 \le \left\|B^Q y\right\|_2,$$

and hence $\|Q\|_2 \le \left\|B^Q\right\|_2 = \varrho(B^Q)$ because B^Q is symmetric and entrywise nonnegative. Finally, extremal eigenvalues and spectral radii of the comparison matrices are bounded by their maximum absolute row sums, proving dominance over the row construction. $\square$

The comparison method requires three symmetric scalar $n \times n$ extremal eigenproblems and does not form the full $dn \times dn$ oADMM matrix.

7 Sparse truncated-kernel certificate

Under periodic boundaries, K_ρ is convolution by a scalar discrete kernel $k_\rho(s)$. Let $\mathcal{R} = -\mathcal{R}$ be a retained finite symmetric set of offsets containing zero, and define $K_{\rho,\mathcal{R}}$ as convolution by $k_\rho(s)\mathbf{1}_{\{s\in\mathcal{R}\}}$. Define

$$\tau_{\mathcal{R}} = \sum_{s\notin\mathcal{R}} |k_\rho(s)|.$$

Let $M_{\mathcal{R}} = C_H(K_{\rho,\mathcal{R}} \otimes I_d)$. Construct the Perron comparison matrices using retained offsets only and let $U_{\mathcal{R}}$ be the resulting affine numerical-range rectangle bound from Theorem 6.1.

Corollary 7.1 (Truncated convolution certificate)**.**

$$\varrho(E_{\rho,\alpha}) \le U_{\mathcal{R}} + \frac{\alpha}{2}\max_i \|C_i\|_2\, \tau_{\mathcal{R}}. \tag{15}$$

Proof. Young's convolution inequality gives

$$\|K_\rho - K_{\rho,\mathcal{R}}\|_2 \le \tau_{\mathcal{R}}.$$

Write $M = M_{\mathcal{R}} + D$. Left multiplication by C_H gives

$$\|D\|_2 \le \max_i \|C_i\|_2\, \tau_{\mathcal{R}}.$$

For a normalized eigenvector x of M with eigenvalue z, $z = x^*M_{\mathcal{R}}x + x^*Dx$. The retained numerical-range rectangle bounds

$$\left|1 - \frac{\alpha}{2} + \frac{\alpha}{2}x^*M_{\mathcal{R}}x\right| \le U_{\mathcal{R}},$$

while $|x^*Dx| \le \|D\|_2$. The triangle inequality yields (15). □

For fixed truncation radius, the comparison matrices are sparse. The kernel can be evaluated by fast Fourier transform (FFT) in $O(n \log n)$ time and extremal comparison eigenvalues can be estimated matrix-free. The bound is rigorous at every radius, including zero; usefulness depends on tightness.

8 Commutator and orientation diagnostics

With $[A, B] = AB - BA$, the resolvent identity yields a direct commutator estimate.

Proposition 8.1 (Cayley commutator bound)**.**

$$\|[C_H, C_G]\|_2 \le \frac{4\rho^2 \|[H, G]\|_2}{(\lambda_{\min}(H) + \rho)^2(\lambda_{\min}(G) + \rho)^2}. \tag{16}$$

Proof. Use $C_H = \mathrm{I} - 2\rho(H + \rho\mathrm{I})^{-1}$ and the identity

$$[(H + \rho\mathrm{I})^{-1}, (G + \rho\mathrm{I})^{-1}] = (H + \rho\mathrm{I})^{-1}(G + \rho\mathrm{I})^{-1}[H, G](G + \rho\mathrm{I})^{-1}(H + \rho\mathrm{I})^{-1}.$$

Multiplying by $4\rho^2$ and bounding the four resolvents by their smallest shifted eigenvalues proves (16). □

For registration,

$$[H, G]_{ij} = (G_0)_{ij}(H_i - H_j),$$

so local coefficient differences control noncommutativity. For the rank-one model and nonzero gradients, set $\widehat{q}_i = q_i/\|q_i\|_2$ and $P_i = \widehat{q}_i\widehat{q}_i^\top$. The sign-invariant orientation difference between two rank-one Hessians is

$$\|P_i - P_j\|_2 = \sqrt{1 - (\widehat{q}_i^\top \widehat{q}_j)^2} = \sin\vartheta_{ij},$$

where $\vartheta_{ij} \in [0, \pi/2]$ is the acute principal angle between the unoriented spans of q_i and q_j. In two dimensions this is also $|\sin(\phi_i - \phi_j)|$, with orientation angles taken modulo π. The projector metric is invariant under either sign change $q_i \mapsto -q_i$. For the controlled study, we additionally compute

$$\chi = \frac{\|HG - GH\|_2}{\|H\|_2 \|G\|_2}.$$

Across the ten cases, χ ranges from 0.012 to 0.239. Larger normalized commutators were generally associated with larger oracle regret; the descriptive Pearson correlation was 0.87. These deterministic systems are designed diagnostics rather than a statistical sample and therefore do not establish a universal relationship.

9 Certificate experiments and scaling

9.1 Structured enclosure comparison

The controlled suite contains 18 cases. Here ϱ_{act} is the true spectral radius at the selected parameters, ϱ_{ora} is the radius at numerically oracle-optimized parameters, and U is the stated rigorous enclosure at the selected pair. The enclosure gap is $(U - \varrho_{\mathrm{act}})/\varrho_{\mathrm{act}}$ and the spectral-radius regret is $(\varrho_{\mathrm{act}} - \varrho_{\mathrm{ora}})/\varrho_{\mathrm{ora}}$. A collapse denotes selection at the boundary of the admissible set.

Table 1: Controlled structured-envelope results; medians over 18 cases.

Method	Enclosure gap	Oracle regret	Collapses
Block resolvent	0.166	0.926	6
Block Gershgorin	0.644	0.560	0
Commutator rectangle	0.602	0.431	0
Comparison rectangle	0.595	0.386	0

Every enclosure contained the exact radius. Minimizing the additive bound produced over-conservative relaxation and collapsed in six of 18 cases. The Perron rectangle retained sign information and had the smallest oracle regret among the signed constructions, but was still too loose for online tuning.

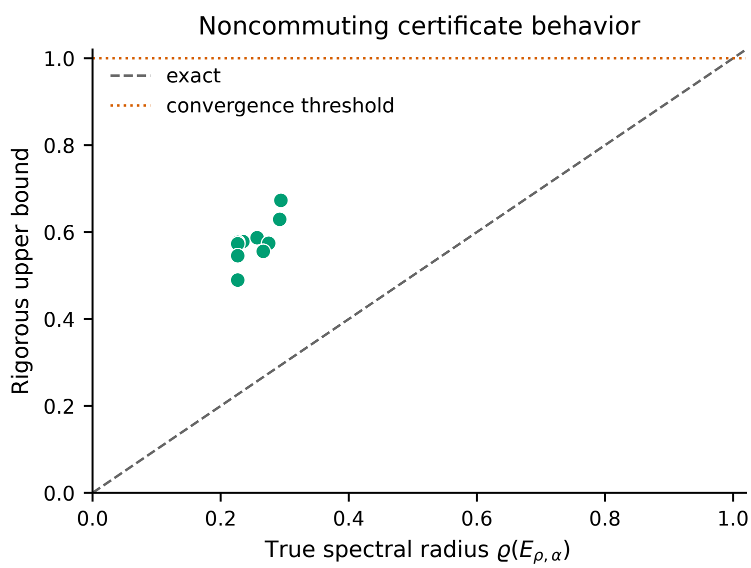

Figure 1: Representative noncommuting certificate values compared with the true spectral radius on the tractable controlled systems. The bounds remain above the exact values but are conservative.

9.2 Controlled variable-coefficient predictor tests

The ten 3×3 cases comprise periodic magnitude-only variation, orientation-only variation, one 90° orientation interface, and three random-gradient stress tests. Let $x_{i,1}$ be the first grid coordinate, $s_i = \sin(2\pi x_{i,1}/3)$, and $e_1 = (1,0)^\top$. The magnitude-only and orientation-only fields are

$$q_i^{\mathrm{mag}} = 0.4(1 + a s_i)e_1, \qquad q_i^{\mathrm{ori}} = 0.4\begin{bmatrix}\cos(a s_i) & \sin(a s_i)\end{bmatrix}^\top,$$

with $a \in \{0.05, 0.2, 0.5\}$ and $a \in \{0.05, 0.2, 0.6\}$, respectively. The interface field has magnitude 0.4 and a $90°$ orientation jump.

The random-gradient cases begin with one fixed periodic Gaussian white-noise field z, apply Gaussian filters K_σ with $\sigma \in \{0.6, 1.0, 1.6\}$ pixels, differentiate, and rescale:

$$q_i^{\mathrm{rnd}} = c_\sigma \nabla (K_\sigma * z)_i, \qquad \left(\frac{1}{N} \sum_i \left\| q_i^{\mathrm{rnd}} \right\|_2^2 \right)^{1/2} = 0.4, \qquad N = 9.$$

The filtering scale describes the precursor scalar field, not monotone smoothness of the normalized rank-one Hessian. These cases vary gradient magnitude and orientation jointly. Every case uses $H_i = 0.05 I_2 + q_i q_i^\top$, the same periodic regularizer with $(\beta, \gamma) = (0.2, 0.1)$, and $\mathcal{I} = [10^{-2}, 10^2]$.

The four-corner predictor attained median spectral-radius regret 0.057 over all cases, with no collapse. Regret was 0.003–0.087 for the single-factor magnitude and orientation constructions, 0.193 for the abrupt interface, and 0.238–0.296 for the random-gradient fields. Their normalized commutators were respectively 0.012–0.139, 0.166, and 0.171–0.239. At the selected parameters, the median absolute relative discrepancy between the four-corner envelope and the true radius was 1.4×10^{-15}, with maximum 2.8×10^{-3}. The latter is a pointwise comparison at $(\rho_{4\mathrm{C}}, \alpha_{4\mathrm{C}})$, not a comparison of the objectives over the parameter domain. It is therefore compatible with nonzero oracle regret: the oracle can attain a smaller true radius at another pair even when the two radii agree closely at the selected pair. The Perron comparison certificate was below one in all ten cases. Its median enclosure gap was 1.29, establishing validity while illustrating conservatism.

9.3 Truncation scaling

The truncated-kernel certificate showed no violation in 20 tests spanning 8^2 through 64^2 grids and five truncation radii. Actual matrix-free spectral radii were approximately 0.269, while certificate values ranged from 0.638 to 0.681, corresponding to relative gaps 1.38–1.54. The most expensive 64^2 retained-radius case took 81.4 seconds on a central processing unit (CPU). A relaxed 128^2 Arnoldi certificate run did not complete within its allocated diagnostic window. The sparse representation scales in principle, but the current eigensolver and bounds are not practical registration-time certificates.

9.4 Additional algebraic verification

Ten thousand randomized curvature configurations were used to compare explicit enumeration of attained modal curvature pairs with the four-corner envelope. No discrepancy was observed. Constant-coefficient periodic systems matched the full reduced spectral radius to numerical precision. These tests validate the implementation of the separable theorem; they do not turn the variable-coefficient predictor into an exact global spectral method.

10 Implementation and baseline details

10.1 Solver

All methods use the same three-level image pyramid, initialization, linearization, stopping tolerances, interpolation, line search, and discrete-Jacobian safeguards. Pixelwise data solves use the Sherman–Morrison formula; regularizer solves use FFT diagonalization. The line search requires objective decrease and a discrete-Jacobian floor of 0.05. When pyramid upsampling violates this floor, only the nontranslational displacement is halved until the check passes. This numerical safeguard does not constitute a global diffeomorphism theorem.

10.2 Baselines

The validation-selected fixed comparator $(\rho, \alpha) = (0.1, 1.0)$ was chosen from $\rho \in \{0.03, 0.1, 0.3, 1, 3\}$ and $\alpha \in \{1, 1.8\}$ using known-deformation cases 1–2 from each of two separate public images. Candidate pairs within 1% of the lowest median mean squared error were ranked by median inner iterations and then by median error. Neither FIRE nor CIMA was used in this selection. Residual balancing starts at $\rho = 1$, updates the penalty from primal and dual residual imbalance, and rescales the dual variable. The Barzilai–Borwein (BB) comparison is a safeguarded penalty proxy with clipping and periodic updates. Fixed oADMM uses $(1, 1.8)$, while Fixed ADMM uses $(1, 1)$. Exact update intervals and safeguards are implemented in `src/registration2d.py`.

10.3 Certificate reference blocks and regularizer symbols

Optional certificate studies may freeze a representative local optical-flow block across space or partition the domain into larger spatial regions. A representative $\bar{q}_r$ may be a center, mean, or robust spatial median, and

$$H_r = \nu I_d + \mu \bar{q}_r \bar{q}_r^\top.$$

For a direct block surrogate,

$$\delta_H = \max_i \left\| \mu q_i q_i^\top + \nu I_d - H_{r(i)} \right\|_2.$$

Periodic or cosine closures provide analytic regularizer endpoints. For an order-p Sobolev regularizer,

$$g(\xi) = \left(\gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^{d} \sin^2(\xi_\ell/2) \right)^p,$$

so

$$g_- = \gamma^p, \qquad g_+ = \left(\gamma + \frac{4\beta d}{(\Delta x)^2} \right)^p.$$

11 Additional benchmark information

11.1 Controlled real-content pilot

A public-domain retinal image and an unrestricted immunohistochemistry image were each subjected to five smooth known deformations. Cases 1–2 selected the single validation-selected fixed comparator; held-out cases 3–5 were not used for tuning and gave six image–deformation tests. Across five order-alternated repeats (30 paired runs), one-shot tuning reduced median inner iterations by 44.4% and median total time by 41.7% (bootstrap 95% interval [40.9, 42.6]%). Selection consumed 1.58% of proposed runtime, median relative mean squared error (MSE) change was -0.0068%, and no tested case had a nonpositive Jacobian.

Table 2: Controlled real-content pilot against the predeclared acceptance gates; medians over paired runs.

Metric	Observed	Gate
Inner-iteration reduction	44.4%	$\geq 15\%$
Total-time reduction	41.7%	$\geq 10\%$
Relative MSE change	-0.0068%	$\leq 1\%$
Selection overhead	1.58%	$< 2\%$
Nonpositive Jacobians	0/6	0

11.2 Coarse direct spectral optimization diagnostic

For each CIMA pair, a variable-coefficient reduced operator was assembled on an 8×8 downsampled grid and its penalty was optimized numerically. At each trial ρ, if $\{\lambda_j(\rho)\}$ are the eigenvalues of $C_H(\rho)C_G(\rho)$, the conditional relaxation was computed from

$$\alpha_\rho \in \operatorname*{arg\,min}_{10^{-6} \leq \alpha \leq 2} \max_j \left| 1 - \frac{\alpha}{2} + \frac{\alpha}{2}\lambda_j(\rho) \right|.$$

The resulting pair was transferred to the full 256^2 registration. The coarse spectral search required a median 97 spectral-objective evaluations and 2.18 seconds of parameter-selection time. Median full-solve iterations were 251, compared with 210 for four-corner tuning, and total times were 4.39 and 2.02 seconds, respectively; target registration error (TRE) was indistinguishable. This diagnostic quantifies repeated spectral-evaluation overhead on a tractable surrogate. Parameter transfer from a coarse operator is distinct from direct spectral optimization of the full-resolution variable-coefficient system.

11.3 CIMA preprocessing and initialization

The public lung-lesion-3 sample comprises five differently stained sections and all ten unordered pairs. Images supplied at 5% scale were converted to a common structural channel by grayscale conversion, contrast-limited adaptive histogram equalization (CLAHE), Sobel-magnitude extraction, Gaussian smoothing, and fixed percentile normalization, and were then resized to 256^2. Manual landmarks supplied at 50% scale were rescaled geometrically and used only for evaluation. Each method used the same image-only robust translation initialization selected from phase correlation, centroid alignment, and clustered scale-invariant feature transform (SIFT) proposals on a 128^2 image.

11.4 FIRE parameter summary

The FIRE result contains 134 pairs and five methods per pair. Percentage reductions are computed pairwise before aggregation. For one-shot four-corner tuning, median $\rho_{4\mathrm{C}} = 0.1008$ (interquartile range (IQR) 0.0397, range 0.0688–0.1402) and median $\alpha_{4\mathrm{C}} = 1.5699$ (IQR 0.1287, range 1.4483–1.6883). No selected $\rho_{4\mathrm{C}}$ reached either endpoint of $[10^{-4}, 10^3]$, and no selected $\alpha_{4\mathrm{C}}$ reached 2. The median h_+ is 0.2249.

11.5 One-shot reuse

On a balanced 18-pair FIRE subset, per-level retuning reduced median inner iterations from 243 to 222 but increased median tuning time from 0.030 to 0.110 seconds and median total time from

2.651 to 3.895 seconds. Accordingly, all headline experiments use one-shot reuse. Retuning at every outer Gauss–Newton iteration was not tested separately.

12 Reproducibility and availability

The experiments used seed 20260806 and `configs/real2d_v1.yaml`. The environment used Python 3.10.9, NumPy 2.2.6, SciPy 1.15.3, pandas 2.3.3, Matplotlib 3.10.9, and scikit-image 0.25.2 on two Intel Xeon Gold 5218 CPUs; the experiments were CPU-only. Timed registrations used one Basic Linear Algebra Subprograms (BLAS)/OpenMP thread. Exact commands and the complete validation sequence are recorded in `results/real2d_v1/reproducibility_log.md`. Raw pair-level outputs, result schemas, and regeneration scripts are stored alongside that log. FIRE preprocessing uses a foreground-percentile-normalized green channel; CIMA uses the structural-channel processing described above. Supplied landmarks are used only for evaluation.

FIRE and CIMA are public datasets subject to their original licenses. Raw FIRE images are not redistributed. Preparation scripts, integrity checks, all nonrestricted pair-level measurements, and the certificate implementations (`src/structured_envelopes.py` and the corresponding `experiments/` scripts) are available at https://github.com/xhxuciedu/admm. The repository includes source code, tests, configurations, processed outputs, and preparation and evaluation scripts.